

(2+1)-Dimensional Chern-Simons Gravity as a Dirac Square Root

Source-backed arXiv sample

Abstract

For (2+1)-dimensional spacetimes with the spatial topology of a torus, the transformation between the Chern-Simons and ADM versions of quantum gravity is constructed explicitly, and the wave functions are compared. It is shown that Chern-Simons wave functions correspond to modular forms of weight 1/2, that is, spinors on the ADM moduli space, and that their evolution (in York’s “extrinsic time” variable) is described by a Dirac equation. (This version replaces paper 9109006, which was garbled by my mailer.)

1 References And Citations

This sample cites source keys [1] and [2]. Section 1 and Equation 1 provide cross-reference coverage.

$$a^2 + b^2 = c^2 \tag{1}$$

References

- [1] Source bibliography entry 1 extracted from arXiv metadata/source key ‘Mart’.
- [2] Source bibliography entry 2 extracted from arXiv metadata/source key ‘HosNak’.
- [3] Source bibliography entry 3 extracted from arXiv metadata/source key ‘Visser’.
- [4] Source bibliography entry 4 extracted from arXiv metadata/source key ‘HosNak2’.